ADITYA UNIVERSITY

(Formerly known as Aditya Engineering College (A))

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PART-I

S.No: 421554

Programe	:	B.TECH
Hall Ticket No:		23A95A3521
Name	:	MADICHARLA MANIKANTA TARUN TEJA
Examination	:	III SEMESTER SUPPLEMENTARY (AR20(R))
Month-Year	:	Nov/Dec 2025
Branch	:	AGRICULTURAL ENGINEERING
Course Code	:	201BS3T12
Course Name	:	Integral Transforms and Applications of Partial Differential Equations
Date of Exam	:	18-11-2025

J. Pavan.

Signature of the Controller of Examination

M.M-Tarun Teja 18/11/25

Signature of the Student with date

NB 18/11/25

Signature of the Invigilator with date

Serial No. Last Page Written	23

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

$f(x) \ldots = \int f(x) e^{ipx} dx \int_0^{\infty} f(x) f(x) e^{-ipx}$

$= \int_0^{1} f(x) \sin px \, dx$ $\quad \sin px \} e^{-ipx}$ $\quad e^{ipx}$

unit 3

$L\{f(t)\} = L\{\cos 3t\} = \frac{s}{s+a} = \bar{f}(s)$

second shifting theorem

$L\{f(t)\} = \ldots \frac{3!}{s+1} = \frac{6}{s} = f(s)$

Multiplication

$L\{f(t)\} = \bar{f}(s) \quad L\{t f(t)\} = \frac{d}{ds} \bar{f}(s)$

division

$L\{f(t)\} = L\{1 - \cos t\} = L\{1\} - L\{\cos t\} =$

Periodic function

$L\{f(t)\} = \frac{1}{1-e^{-s\pi}} \int_0^{\pi} e^{-st} f(t) \, dt$

improper integral by

$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \, dt = \bar{f}(s)$

unit 4

$= L\{f(s-a)\} = e^{at} L^{-1}\{\bar{f}(s)\}$

$= L^{-1}\{f(s-a)\} \ldots$

partial fraction

$\frac{\ldots}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$

$\frac{s^2}{(s+1)(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+1} + \frac{C}{s+2}$

Polynomial

$\frac{s}{s+s^4} = \frac{As+B}{s^2+s+1}$

$= \frac{s^2}{(s+1)(s+2)} = \frac{As+B}{s+1} + \frac{Cs+D}{s+2}$

$L^{-1}\{\bar{f}(s)\} = f(t)$

Q.No.

unit-1

1a) fourier series of $f(x) = \begin{cases} x & \text{if } 0 < x < 3 \\ 6-x & \text{if } 3 < x < 6 \end{cases}$

given that,

$$a_0 = \frac{1}{\pi} f(x)\, dx \quad ①$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\, dx \quad ②$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx \quad ③$$

$$a_0 = \frac{1}{\pi} f(x)\, dx$$

$$= \begin{Bmatrix} x & 0 < x < 3 \\ 6-x & 3 < x < 6 \end{Bmatrix}$$

$$= \left\{ \frac{x}{6} + \frac{x}{3} + \frac{x^3}{3} \right\}$$

$$= \left\{ \frac{x}{3} - \frac{x}{6} \right\}$$

$$= 3 \times 6 \quad = 18$$

$$f(x)\, dx = \int_{-6}^{1} \{ 0 < x < 3 \} \{ 6-x \; 3 < x < 6 \}$$

$$f(x)\, dx = \begin{Bmatrix} 0 & < & z \\ 3 & x & 3 \end{Bmatrix}$$

$$f(x) = \frac{(x-6)}{6}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\, dx$$

$$= \frac{1}{\pi}\int_{0}^{\pi} (\pi - 0)\,(\pi - < 3)\,(\pi - 6)$$

$$= \frac{1}{\pi}\int x^3 - 3)\,(x^2 - 6)$$

Q.No.

$= 0[-\pi+\pi) = 0$

$= \begin{cases} x & 0 < x < 3 \\ 6-x & 3 < x < 6 \end{cases}$

= The interform fourier series transform

$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx \, dx$

$b_n = \frac{1}{\pi} \frac{0}{3} \times \frac{3}{6}$

$= 0$

$= \frac{1}{\pi} f(x) \sin nx \, dx$

interval is called

$= [-1-\pi]^2 [-(x-2)^2 - (x<3)$

$= \frac{1}{2\pi}\{-\pi^2 - \pi^1\}$

$= \frac{1}{\pi}[x^2 - \pi^1]$

The ~~integrive~~ interval is fourier series

$= \frac{1}{\pi}\int^{\pi} f(x) \sin nx \, dx$

$b_n = [3+6]^2$

$= 81$

$\boxed{b_n = 81}$

$f(x) = \{ b_n = \frac{1}{\pi}\int_0^{\pi} \cancel{0.66}\ 0.81\%$

Q.No.

1b). $\cdot f(x)$ $x \cos x$ interval $(-\pi, \pi)$.

given that $a_0 = \frac{1}{\pi} f(x)\, dx$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \cos nx\, dx$$

$$b_n = \frac{1}{\pi}\int_{\pi}^{\pi} f(x) \cdot \sin nx\, dx.$$

$$a_0 = \frac{1}{\pi} f(x)\, dx$$

$$x \cos x \; f(x)^{-2}$$

$$= [-\pi, \pi)^{0} - [-\pi+\pi)^{2}$$

$$= f(x)\, dx \quad (-\pi, \pi)$$

$$f(x) = x \cos x \; nx$$

$$x = \frac{1}{\pi} f(x)\, dx$$

$$a_0 = \frac{1}{\pi} f(x)\, dx$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{\pi} - \frac{\pi}{2}\right]$$

$$= \frac{1}{\pi}\left[-\frac{\pi}{\pi} \cos nx\right]$$

$$a_n = \frac{1}{\pi}\int_{-}^{\pi} \cos nx\, dx$$

$$f(x) = x \cos x$$

$$= \int_{0}^{\pi} \cos x\, dx$$

$$= x \cos(x)$$

$$= 3.0711$$

$$b_n = \frac{1}{\pi}\int_{\pi}^{\pi} f(x) \sin nx\, dx$$

$$\cos nx$$

$$= x \cos nx \cdot f(x)\, dx$$

$= f(x) \cos x \rightarrow ①$

$= \cos x \,(-\pi; \pi))$

$= b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin dx$

$= \cos nx \, dx$

$= \frac{1}{\pi}\int_{\pi}^{\pi} \frac{1}{\pi}\int^{\pi} f(x) \sin dx$

$= \cos \frac{x^2+1}{0}$

$= x \cos nx = \int_{-\pi}^{\pi} (\cos nx + 1))$ //.

unit-II

3a) fourier integral, $\int_0^{\infty} \frac{\cos \lambda x}{1+\lambda^2} d\lambda = \frac{\pi}{2} e^{-x}, x > 0$..

given that:-

$= f(x) = \int_{-\infty}^{\infty} \sin px \int_{-\infty}^{\infty} f(t) \sin pt \, dt \, dp$ — ①

$= f(x) = \frac{2}{\pi}\int_{-\infty}^{0} \sin x \int_0^{\infty} f(t) \sin pt \, dt + dp$

$\int_0^{\infty} \frac{\cos \lambda x}{1+\lambda^2} d\lambda = \frac{\pi}{2} e^{-x}, \ x > 0$ ..

$= f(x) \frac{\cos \lambda x}{1+\lambda^2} d\lambda \frac{\pi}{2} \ x > 0.$

$= f(x) = \frac{2}{\pi}\int_0^{\infty} \cos x \int_0^{\infty} f(t) \sin px + 0$

$= f(x) = \frac{2}{\pi}\int_{-0}^{\infty} \sin x \int_0^{\infty} f(t) \cos pt \, dt$

$= \frac{\cos \lambda x}{1+\lambda^2} d\lambda = \frac{\pi}{2} e^{-x}$

fourier integral $\int_0^{\infty} \frac{\cos \lambda x}{1+\lambda^2} d\lambda = \frac{\pi}{2} e^{x} x > 0.$

Q.No.

$$f(x) = \frac{2}{\pi}\int_0^{\infty} \cos x \int_0^{\infty} f(t) \cos x\, pt\, dt$$

$$f(x) = \frac{2}{\pi}\int_0^{\infty} \sin x \int f(t) \sin pt\, dp$$

$$\int \frac{\cos \lambda x}{1+\lambda^2}\, d\lambda\ \frac{\pi}{2}\quad x > 0.$$

$$f(x) = \frac{e}{\pi}\int_0^{\infty} \cos x \int_c^{\infty} f(t) \cos nx\, dt.$$

$$= \int_0^{\infty} \cos x \int_0^{\infty} f(t) \cos nx\, dt$$

$$= \int_0^{\infty} \frac{\cos \lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2} e^{-x}\quad x > 0.$$

$$= \int_0^{\infty} \sin px \int^{\infty} f(t) \sin t\ \frac{\pi}{2} e^{-x}\quad x > 0 \ldots$$

$$= \int_0^{\infty} \frac{\cos \lambda x}{1+\lambda^2}\, d\lambda = \frac{\pi}{2} e^{-x} - 0$$

$$= \frac{\cos \lambda x}{2+\lambda^2}\quad \frac{\pi}{2} e^{-x}.\ 1)$$

3b) given ~~data~~ that,

$$f(x) = \int_{-\infty}^{\infty} \sin px \int_{-\infty}^{\infty} f(t) \sin pt\, dt$$

$$f(x) = \begin{Bmatrix} 1 & 0 < x < 2 \\ 0 & \text{if } x > 2 \end{Bmatrix}$$

$$f(x) = \begin{Bmatrix} 1 & 0x < 2 \\ & x > 2 \end{Bmatrix}$$

$$= f(x) = \int_0^{\infty} f(x) = px) \}$$

$$= \begin{Bmatrix} 1 & 0 < x < 2 \\ 0 & x > 2 \end{Bmatrix}$$

$$= 2 \times 2 \quad = 4$$

Q.No.

$= \int^{Le} x^2 \; |x| < a$

$|x| >$

$f(x)$ sin $\frac{\{1+x\}}{= 89}$ $\{2+x\}$ $= 90$

$\left\{\frac{x^2}{x^2} \frac{<}{2}\right\}$

$= f(0) = \{f(x) = \{\frac{\sin x}{x}\}_1^0$

$= f(x) = \begin{cases} 1 & 0 < x < 2 \\ 0 & x > 2 \end{cases}$

$= \frac{0+1+2}{3}$

$= f(x) = \sin x$ cosine

$f(x) = 0.3\%$

unit-III

5a) find $L\{\int_0^t e^{-t} \cos t \, dt\}$

$L\{f(t)\} = L\{\cos t\} = \frac{s}{s+a} = F(s)$

using multiplication

$L\int_0^t e^{-t} \cos t \, dt$

$= \frac{\cos t \, dt}{0}$

$= \int_0^t e^{-t} \cos t \, dt$

$= \frac{s}{s+a} = F(s)$

Q.No.

$L\{\int_0^t e^t \cos t \, dt$

$= L\{\cos dt\} f(t))$

$= \frac{\cos t \; dt}{0}$

$= \int_0^t e^{-t} \frac{\cos t}{x} f(t))$

$= \int_0^t \cos t \; dt \; \overline{\;}eipx$

$= \int_0^t e^{-t} \cos t \; dt\}$

$= \{e^{-t} \cos t \; dt\}$

$= \frac{\cos dt}{e^{-t} dt}$

$= L\{\cos t dt\} = \frac{s}{s+a}\}_0^0$

$= L\{f(t) = L\{\cos dt\}$

$= \frac{\cos dt}{0} = \frac{\sin dx}{\cos nt}$ //.

5b) given that,

$L\{f(t)\} - L\{-\cos t\} \; L\{1\} - L\{\cos t\})$.

$f(t) = \begin{cases} K \text{ if } 0 < t < a \\ -K \quad a < t < 2a \end{cases}$

$f(t) \; \frac{2a + 0K}{-K + 22a}$

$f(t) = L\{-\cos t\} \quad L\{1\} - L\{\cos t\})$

$f(t) = \int_{-0}^{\infty} \cos t - \frac{d}{dt} \cdot \{1\}\} - L\{\cos t\}$

$= \{f(t) = \frac{\cos t}{n} \cdot c \; \frac{\cos t}{n} \; \{\{t\}\} \; L\{\cos nx\}$

Q.No.

$$= f(t) = \left\{ \frac{1}{t^2} \quad \frac{0\, t < a}{0 < 2a} \right\}$$

$$= f(t) = (0 < 2a \cdot < a)$$ //.

unit - IV

7a) $L^{-1}\left\{ \frac{2s-3}{s^2+4s+13} \right\}$ given that,

$$= L\{f(s-a) = e^{-at}\, L^{-1}\{(s)\}^{-1}$$

$$= L\{f(s-a) = e^{at}\, L\{\bar{f}(s)\}$$

$$L^{-1}\left\{ \frac{2s-3}{s^2+4s+13} \right\} = 0$$

$$L\left\{ \frac{2s-3}{s^2+4s+13} \right\} = 0$$

$$L^{-1}\left\{ \frac{2s-3}{s^2+4s+13} \right\}$$

$$L^{-1}\left\{ \frac{s^2+4s+3}{0} \right\}$$

$$= L^{-1}\left\{ \frac{s^1+4s-3}{s^2+4s+13} \right\}$$

$$= L^{-1}\left\{ \frac{2s-3}{s^2+4s+13} \right\}$$

$$L\left\{ \frac{f\,2s-3}{s^2+4s+13} \right\}$$

$$L\left\{ \frac{f\,2s-3}{(s+0)(s+1)(s+2)} \right\}$$

$$L\left\{ \frac{f(s)23}{(s+0)\,(s+1\quad s+2)} \right\}$$

Q.No.

$$= L\left\{\frac{2s-3}{(s^2+4s+13}\right\}$$

$$= L\left\{\frac{2s-3}{s^2+4s+13}\right\}$$

$$= L^{-1}\left\{\frac{s+1}{s+2}\right\}$$

$$= \frac{L+1}{L+2}$$

$$= \frac{s^2+4s+13}{0}$$

$$= L\{f(t)\ (s-a)\ = e^{-at}\ L\{f\ (s)\}$$

$$= f(t) = 0\quad L\{F(s)\}$$

$$\left\{\frac{2s+3}{s^2+4s+13}\right\}_{//}$$

7b. given that:-

$$\frac{s}{s+1\ (s+2} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$L^{-1}\left\{\frac{s^2}{(s+4)\ (s^2+9)}\right\}$$

using theorem, $L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$

$$= L^{-1}\left\{\frac{s^2}{(s^2+9)\ (s^2+9)}\right. = F\ (s)$$

$$= \left\{\frac{s}{s+1\ (s+2)}\right\} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$= \frac{s^2}{(s+1)(s+2)\ (s+3\ )} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}$$

Q.No.

$$= \left\{ \frac{s^2}{(s^2+4)(s^2+9)} \right\} = F(s)$$

$$= \left\{ \frac{L^{-1}}{(s^2+4)\ (s^2+9)} \right\} = 0$$

$$= \left\{ \frac{s^2}{(s^2+4)(s^2+9)} \right\}$$

$$= \frac{s+2}{(s^2+1)\ (s^2+2)} = \frac{As+B}{s+1}\ \frac{B+D}{s+2}\Big)$$

$$= \left\{ \frac{s^2}{(s^2+4)(s^2+9)} \right\} = 0$$

$$\frac{s+2}{(s^2+1)\ (s+2)^2} = f(x) = 0$$

$$= L^{-1}\left\{(s^2+4) + (s^2+9)\right\} = \frac{A}{s+1}\ \frac{B}{s+2}\ \frac{C}{s+3} = \frac{Cs+D}{\cancel{Cs+D}}$$

unit-V

9a) given that,

$$\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u \quad (x,0) = 0$$

let $(x,t) - 0$

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial t} - 0$$

$u(0,y) = 6e^{-3x}$

$u(x,y) = X(x)\ T(t) \Rightarrow 3$

$$\frac{\partial u}{\partial x} = X'(x)\ T(t) \to 0$$

$$\frac{\partial u}{\partial y} = X(x) + (y) \} - 4$$

Q.No.

$6x'(X)_1 (g)(t) + y'(y = 3X(X)\, x(y)$

$6x'(X = 3x(X)\, y)_2 = X(x)\, y(y)$

$(\cancel{6}D-\lambda)X) = 0$

$4M-\lambda = 0$

$4M = \lambda$

$M = \frac{\lambda}{4}$

$x(y) = c_1 e_{1,-}$

$\frac{\cancel{6}(y)\, X(X) = \lambda}{y(y)}$

$3Y(y) - Y'(y) = \lambda y(y)$

$3y(y) - y'(y)' = \lambda y(y = 0$

$-3y(\lambda-4)\, 3(3-\lambda) = 0$

$(D-(3\lambda)\,\Lambda(y) = 0$

$M = (6-\lambda) = 0$

$\frac{\partial u}{\partial x} = 2\frac{\partial y}{\partial t} + u(x_{10}) =$

(x,t) — ①

$3Y(y - Y(y) = \lambda y(y)$

$\cancel{6}$ $-3x$ using by varibies.

10a) given that..

$\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial t} + u(x_{10}))$

let (x,t) — ①

$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial t}$ — ①

$u(0,y) = 8e^{-3y}$

Q.No.

$u(0,y) = 8e^{-3y} - 3y$

$u(x,y) = x'(x)\ \pi(t)$ — ①

$\frac{\partial u}{\partial x} - x'(x) y(y)$

$\partial y = x\ (x) y(y)$ } — ④

$8x'(x) x(y)\ (t)\ x y'\ (y=3\ x(x)\ x(y))$

$8x'(x) x = 3x = (x) = x(x) y(y)$

$(8e^{-3y}) = 0$

$(4D - \lambda) = 0$

$4m = \lambda = 0$

$4m = \lambda$

$m = \frac{\lambda}{4}$

$x(y) = c_1 e_{11}$

$\frac{8(y)\ y(x)}{yx} = \lambda$

$8y(y) - y'(y) = \lambda(y)$

$8y(y) - y'(y)' = \lambda y(y=0)$

$8y\ (\lambda = 4) \cdot 8\ (3-\lambda) = 0$

$(D = (3-\lambda)\ x(y)) = 0$

$m = 8$ ①

$8y - e x(0,y)$ let $(u,0)$

$\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial y} = 0$

Q.No.

$\lambda = -4$ | $\lambda = 3$

$D = -1$ | $D\lambda = 8$

$\boxed{\lambda = 1}$ //.

9b) given that.

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$u(0, y) = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$\partial t^2 = c^2 \frac{\partial^2 y}{f x^2}$$

$$\frac{f(2x)}{f(x)} = \frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 + y}{\partial x^2}$$

wave equation,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 t}{\partial x^2}$$

$$f(0)\,t = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$= u(0, y) = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$t \frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$= \frac{\partial^2 y}{\partial x} + \frac{\partial y}{\partial x} - 3y = e^t$$

$$= y(0) = y'(0) = 1.$$

Q.No.

$$= \frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

$$c(0,2) = (c\cancel{t}\, 0.3)$$

$$= c^2 \frac{\partial^2 y}{\partial x^2}$$

possible solutions

$$= c(x^2) \frac{\partial^2 y}{\partial x^2} //.$$

6a) given that,

formula $L\{f(t)\} - L\{1 - \cos t) \; L\{1\} - L\{\sin t\}\}$

$t^2 e^{-3t} \sin 2t$

$f(x) = L\{1 - \cos t) \quad L\{1\} - L\{\sin t\}\}$

$= \sin 2t + t$

$= L\{f(t)\} \; (-\cos t) \; L\{1\} - L\{\sin t\}.$

$$= \frac{\sin 2t \;\; -3\cos t}{0}$$

$$= L\{1\}\left\{\frac{\sin t}{\cos t}\right\}$$

$= \{t^2 \cdot \sin 2t\}$

$f(0) = L^{-1}\{t^2 \sin 2t\}$

$$= \cancel{t0} \; \frac{\cos t}{2} + \frac{\sin t}{2}$$

$$= L\{-\cos t\} \quad L\{1\} \quad \frac{\sin 2t}{0}$$

Q.No.

$= \int_0^t e^{-t} \cos t \, dt$

$= \int e^{-t} \cos t \; dt$

$= \int_0^{\pi} e^{-t} \cos t \; nt$

$= \int_0^{\pi} \frac{e^{-t} \cos t \, nt}{0}$

$= \int_0^t - L\{f(x)\} \cos t \; L\{1\} - L\{\sin nt\}\}$

$= \int_0^t f(x) = \sin 2t = 0$

$= \int_0^t f(x) \quad e^{-3t} \sin t \; nt \; dt \}$

$= \{f(x) = e^{-3t} \sin t \; nt \; \cos t \; nt\}$

$= L^{-1} \{t^2 e^{-3t} \sin 2t\}$

$= L\{f(t)\} = L\{L - \cos t \; \{L(1)\} \; \frac{\sin t}{2t}$ //-

6 b) given that;

$$\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} \, dt$$

$L\{f(t)\} = \frac{1}{1 - e^{-st}} \int^{\pi} e^{-st} f(t) \, dt$

$L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) \; dt = F(s)$

$= \frac{\cos 6t - \cos 4t}{t} \, dt$

$= \frac{\frac{\cos 6t - \cos 4t}{t}}{0} \, dt$

Q.No.

$L\{f(t)\} = \frac{\cos 6t - \cos 4t}{t}\, dt$

$\frac{\cos 6t - \cos 4t}{t}\, dt$

$f(t) = f(t)\ \frac{\cos 4t}{t}\, dt$

laplace transforms

$\int_0^\infty \frac{\cos 6t - \cos 4t}{t} \cdot f(t)$

$\frac{\cos 6t \quad - \cos 4t}{0.42 \quad 8}$

$L\{f(t) = \int_0^\infty \frac{\cos 6t - \cos 4t}{t}$

Now $\frac{\cos 6t - \cos 4t}{4t}$

$= L\{\int_0^\infty \frac{\cos 6t}{t} - 1\}$

$= L\{\int_0^\infty \frac{\cos t}{t} + 1\}$

$= L\{\int_0^\infty \frac{\cos 6t - \cos 4t}{t}\}_0^1$

$= L\{\int_0^\infty \frac{\cos^2 t - \cos 4t}{t}\}_0^1$

$= L^{-1}\{\int_0^\infty \frac{\cos 6t - \cos 4t}{0}\}$

8a) given that,

$$L^{-1}\left\{\frac{S}{(S^2+a^2)^2}\right\}$$

$$\frac{S}{S+S4} = \frac{As+B}{S^2+S+1}$$

$$= \frac{S+2}{(S^2+1)}(S^2+2) = \frac{AS+B}{S^2+B}\ \frac{B+D}{S+2}$$

$$L^{-1}\{f(s)\} = f(x)$$

$$\frac{S+3}{(S^2+1)}(S^2+3) = \frac{AS+B}{S^2+B}\ \frac{D+B}{S+a^2}\}$$

$$= \frac{S}{(S^2+a^2)} = Q1$$

$$= \frac{S}{S+S2} = \frac{As+B}{S^2+S+1}$$

$A+B=1$ | $BC+CD=1$ | $\frac{CD+SD}{S^2+B+S+2}$ | $D=0$

$AB+CD=1$ | $DC+DF=1$ | S^2+a2 | $D2=a^2$

$B=0$

NOW,

$$=\left\{\frac{S}{(S^2+a^2)^2}\cdot\frac{AS+B}{(S^2+a^2)}\right.$$

$$=L^{-1}\{(S+1)(S+2) = \frac{A}{S+1}+\frac{B}{S+2}+\frac{C}{S+3}$$

$$=\left\{\frac{S}{S^2+a^2}\right\} = L^{-1}\left\{\frac{S}{a^2+a^2}\right\}$$

Q.No.

$$\left\{\frac{s}{(s^2+a^2)^2}\right. = f(0)$$

$$= \frac{s}{s+4} = \frac{As+B}{A^2+B} = CB + D$$

$$= (s^2+a^2)^2 + (b^2+c^2)\ (Ds+CD)\ ,,$$

$$= \left\{\frac{s}{(s^2+a^2)^2}\right\} = f(0)$$

$$= (s^2+a^2)^2 = 0$$

$$= \frac{(s^2+a^2)}{(s^2+b^2)}\ \frac{(s^2+c^2)}{(s^2+D^2} = L\{s^2+a^2\}$$

$$= L^{-1}\left\{\frac{s}{s^2+a^2}\right. \ u = 0,,$$

$$= \frac{s}{s^2+a^2} = \bar{f}(B) = (s)\ ,,$$

8b) given that,

$$L^{-1}\ \frac{\partial^2 y}{\partial x^2} + 4\frac{\partial y}{\partial x} + 3y = e^{t},\quad y(0) = y'(0) = 1$$

$$L^{-1}\{f(x)\} = f(x)\ 3y = e^{t}$$

let $y(0) = y'(0) = 1$

$$\frac{\partial^2 y}{\partial x^2} + 4$$

let, $(u, 0)\ \frac{\partial^2 y}{\partial x} + 4$

periodic functions,

Q.No.

$$\frac{\partial^2 y}{\partial x^1}+4\frac{\partial y}{x}+3y=e^{3t}$$

$$y(0)=y'(0)=1$$

$$=\frac{\partial y}{\partial t}+3y=e^{-3t}$$

$$=\frac{\partial^2 y}{\partial x^2}+4\cancel{\partial x}=0$$

$$=\frac{\partial^2 y}{\partial x^1}+4=c^{s\cancel{c4}}=0$$

$$=\frac{\partial^2 y}{\partial x^2}+4\frac{\partial y}{\partial t}+3y=e^{-t}$$

$$=\frac{\partial^2 y}{\partial x^2}+4\frac{\partial y}{\partial t}\ 3y=e^{-t}$$

$$=\frac{3\times 4}{12}$$

$$\frac{12}{3}$$

$$=4$$

$$\frac{\partial x^2}{\partial x^2}+4\frac{\partial y}{\partial x}+3y=e^{t}$$

$$f(x)=\frac{\partial x^2}{\partial x^2}+\frac{\partial y}{\partial x}+3y=4_{//}$$

Unit-5

10b). given that

$$\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}.$$

$$u(x,0)=3\sin n\pi x,$$

Q.No.

$u(0,y) = X(x)\,T(t) > 0$

$u(0,t) = 0 \quad u(1,t) = 0$

$0 < x < 1, t > 0.$

$u(x,0) = u(0,t) = 0$

$3 \sin n\pi x \quad u(t) = 1.$

$= 3 \sin n\pi x \quad u(0,t) = 0$

$= u(1,t) = 0$

$= 0 < x < 1, t > 0$

$= 3 \sin n\pi x$

$u(0,t) = 0 \quad (1,t) = 0$

$$\partial u(x,0) = \frac{3 \sin n\pi x}{0}$$

$u(0,t) = 0$

$$3 \sin n\pi x \quad \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} = 0$$

$D = \lambda = 0$

$u(0,y) - 3 \sin n\pi x$

$$= \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} = 0$$

$= (1,t) = 0.$

$= u(0,y) = 3 \sin n\pi$

$(0,y) \quad (0,t) \quad (1,t) = 0$

Q.No.

$u(x,0) = 3 \; (\sin n\pi x = 0)$

$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} = \text{with } u(1,t)$

$= \frac{\partial u}{\partial t} \frac{\partial x^2}{(1,t)} = 0$

$= 3 \sin n\pi x - u(x-0)$

$3y(y) - y'(y) = d(y)$

$= (3 (D-d) \, x(y) = 0$

$= \frac{\partial u}{\partial t} \frac{\partial^2 u}{\partial x^2} \cdot u(1,t)$

$= u(x,0) \cdot$ let,

$= x(\sin nx) + \sin nx \; inx$

$= \cancel{u(0,t)} = \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$

$= (t > 0)$

$u(x,0) = 3 \sin n\pi x$ //.

Q.No.

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SEMI-LOG PAPER (5 CYCLES X1/10")